
UNIVERSITY OF WARWICK

SECOND YEAR PHYSICS/ASTRO LABORATORY

Experiment A3: Radio Astronomy

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Aims

To demonstrate the value of observations of the Universe outside the optical range.

To learn the elements of radio astronomy.

To evaluate the different impacts of random and systematic uncertainties.

To engage with cutting-edge research literature.

Objectives

To measure the beam-width of a radio telescope from observations of the Sun

To analyse 21-cm emission data from hydrogen in our Galaxy

To use the 21-cm emission to plot the rotation curve of the inner part of the Galaxy

To interpret results in the context of published research.

Please read through the script prior to the experiment. Think about what details you want to record. You should write the answers to the following questions in your notebook before starting. A demonstrator will ask you about them.

- Q1) What is the physical origin of 21-cm radiation?
- Q2) Why does 21-cm emission only give the rotation curve of the Galaxy at galactocentric radii less than that of the Solar system?
- Q3) What mass is required to keep the Sun in a circular orbit at a speed of 220 km s^{-1} 8 kpc from the centre of the Galaxy?
- Q4) What is the angular resolution of the 76-m diameter Lovell Telescope at Jodrell Bank for a wavelength of 21-cm?

Checkpoint 1 - consult the demonstrator when you have answered the questions above. This consultation may need to occur after the initial briefing.

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Statutory and advisory notification of HAZARDS

Experiment: A3 Radio Astronomy

Electrical hazards: AC mains to equipment. No hazard if used in accordance with instructions.

Chemical hazards:

Mechanical hazards:

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Experiment A3 Radio Astronomy

1 Introduction

In this experiment, measurements of 21-cm emission from hydrogen gas will be used to map the rotation of our Galaxy. Our Galaxy, the Milky Way, is a gravitationally-bound structure of stars, gas and dust. Although we see it edge on, we know that the Galaxy comprises sub-structures including a rotating disk, in which our Sun is embedded, and a central bulge in which most of the light of the Milky Way is concentrated.

Hydrogen is the most abundant element in the Universe, making up about 80% of the mass of normal “baryonic” matter. In our Galaxy large amounts of hydrogen are locked up in stars but much also resides in the space between the stars, the “interstellar medium” or ISM. Hydrogen in the ISM exists in three forms: molecular hydrogen, H_2 , familiar to us from hydrogen gas on Earth, ionised hydrogen (protons and electrons), and neutral hydrogen also known as HI (pronounced “aitch-one”).

This experiment primarily concerns neutral hydrogen, HI, in which hydrogen is in its atomic as opposed to molecular form. This is favoured by the low densities in the ISM, which is emptier than the best vacuums on Earth. The temperatures in the ISM are low, ~ 100 K, and so the majority of neutral hydrogen exists in its ground state with the electron in its lowest energy level.

If this was all there was to HI, it would be extremely hard to detect. Fortunately, the ground state of hydrogen is actually split into two very closely-spaced levels owing to magnetic interaction between the proton and electron (hyperfine splitting). The spins of the electron and proton can either be parallel or anti-parallel. The parallel configuration has a slightly higher energy than the anti-parallel one (think of bar magnets), and when a hydrogen atom flips from the parallel to the anti-parallel state, a photon is emitted at a wavelength of 21-cm or equivalently a frequency of $\nu = 1.4204058$ GHz corresponding to the difference in energy, $E = h\nu$. The energy splitting of the ground state is so small that atoms can easily be excited back to the parallel state by collisions, even in the cold conditions of the ISM, so that clouds of neutral hydrogen are constantly emitting 21-cm radiation throughout our Galaxy and other galaxies.

21-cm radiation was predicted by van de Hulst in 1944 and first detected in 1951 by Ewen and Purcell (1951). The location of 21-cm radiation in the radio part of the electromagnetic spectrum means that it is unaffected by the interstellar “dust” that completely blocks our view of most of our Galaxy at optical wavelengths. In addition, this means that we can observe it at any time of the day or night, because although the Sun is easily seen at radio wavelengths, it does not swamp all other sources as it does during the day at optical wavelengths. The radio sky looks *completely* different from the sky we are used to from our eyes. Modern astronomy makes great use of observations beyond the confines of optical light.

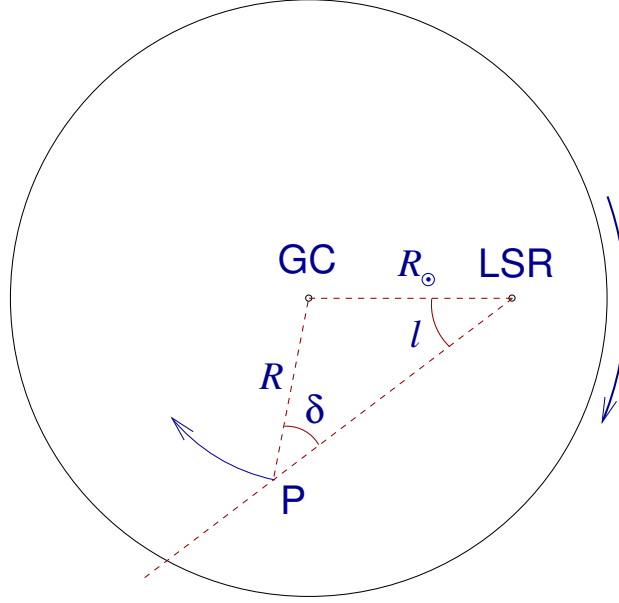


Figure 1: A schematic view of the Galaxy for an observer sited in the direction of the North Galactic Pole showing the definition of Galactic longitude, l , defining the direction of a point P as seen from the Solar system.

2 Theory

21-cm radiation has a very precise frequency, but any motion of the source relative to Earth will shift the frequency because of the Doppler effect:

$$\nu' = \nu_0 \left(1 - \frac{v_R}{c} \right), \quad (1)$$

where ν_0 is the rest frequency, ν' the observed frequency, c is the speed of light and v_R is the “radial velocity”, i.e. the component of velocity along our line-of-sight, defined to be positive if a source recedes from Earth. This can be used to measure the radial velocities of HI clouds in the beam of the radio telescope.

Most stars and gas in our Galaxy are located within a thin disk where they execute nearly circular orbits around the Galactic centre (Fig. 1). Since the Sun is also located inside the disk, the disk appears as a band running around the entire sky, cutting it in half, the “Milky Way”, much like the equator divides the Earth into two halves. In the same way that longitude measures position along the equator of Earth, so does “Galactic longitude” l measure the position along the Milky Way. Galactic longitude is defined to be zero at the Galactic centre (GC in Fig. 1).

Assume that the circular orbital speed at radius R is given by $V(R)$. Then the radial velocity of point P in Fig. 1 relative to the “Local Standard of Rest” (LSR), a point near the Sun moving on an exact circular orbit around the Galactic centre, is given by

$$v_R = V(R) \sin \delta - V(R_\odot) \sin l. \quad (2)$$

Using the sine rule

$$\frac{\sin \delta}{R_\odot} = \frac{\sin l}{R} \quad (3)$$

to substitute for $\sin \delta$ gives

$$v_R = R_\odot \sin l \left(\frac{V(R)}{R} - \frac{V(R_\odot)}{R_\odot} \right). \quad (4)$$

The fractions inside the parentheses are the angular speeds at the respective radii e.g. $\Omega(R) = V(R)/R$. It is safe to assume that the angular speed rises as one approaches the Galactic centre. Thus for $0 < l < 90^\circ$, the radial velocity reaches a maximum at the closest point along the line-of-sight to the Galactic centre where $\delta = 90^\circ$ and $R_\odot \sin l = R$. Therefore

$$v_R(\max) = V(R) - V_\odot \sin l, \quad (5)$$

where $V_\odot$ is shorthand for $V(R_\odot)$. Thus, if we can measure $v_R(\max)$ as a function of Galactic longitude l , we can deduce the rotation speed as a function of galactocentric radius:

$$V(R) = v_R(\max) + V_\odot \sin l, \quad (6)$$

where $R = R_\odot \sin l$. For $270 < l < 360^\circ$, one replaces $v_R(\max)$ by the minimum velocity $v_R(\min)$; for $90 < l < 270^\circ$, the analysis breaks down because the line-of-sight no longer includes the point of closest approach.

To apply Eq. 6 one requires values for $R_\odot$ and $V_\odot$. One way to measure the distance of the Sun from the centre of the Galaxy is from observations of stars orbiting the supermassive black-hole located there (known as Sgr A*). This gives $R_\odot = 7.9 \pm 0.4$ kpc (Eisenhauer et al., 2003). The Sun's orbital speed can be measured by studying the motions of stars and the apparent change in position of Sgr A*. This gives $V_\odot = 220 \pm 13$ km s⁻¹ (Eisenhauer et al., 2003).

Once one has a rotation curve then the mass within radius R , $M(R)$, can be estimated by balancing gravitational and centripetal accelerations:

$$\frac{GM(R)}{R^2} = \frac{V^2(R)}{R}. \quad (7)$$

Note that mass outside radius R has no effect on the orbital speed at R , due to the gravitational equivalent of Gauss' law in electrostatics. It is the measurement of rotation curves of other galaxies compared to the distribution of visible matter that provides some of the strongest evidence for the existence of "dark matter" in the Universe. For much more information on this and motions within the Galaxy, see Carroll and Ostlie (2006)

3 Experiment

3.1 Learning Arrangements

In this experiment, you will **take your own data** from one of the five 2.3m-diameter radio telescope dishes located on the roof of the Physics building. This will be **combined with archival data** to create a rotation curve for the galaxy.

Associated files, videos and archival data are linked from the course Moodle page at <https://moodle.warwick.ac.uk/mod/page/view.php?id=2153041>.

You and your lab partner will each be expected to acquire **three** or more measurements through direct use of a shared telescope, and these will be supplemented with archival measurements acquired by other observers. You will need to read these instructions thoroughly

and carefully before starting the experiment. You should ensure that **all** experimental details are fully recorded in your lab record: what you're doing, why, what problems you encounter and how you solve them, data and analysis filenames, telescope and settings used, the outcomes of discussions with your partner and the demonstrator, etc.

A briefing will be delivered in person at the start of the first session. Please be ready for this to start promptly. You should plan to take notes during this briefing.

3.2 Measuring radio power

The first part of the experiment involves measuring the total power emitted by the Sun in a narrow range of radio frequencies, in order to evaluate a key property of the telescope.

Each radio telescope is connected to a single PC in the lab and is controlled from there. The equipment measures the power received in narrow ranges of frequency (the telescope's *bandpass*). The central frequency can be rapidly tuned to result in a series of points (channels) representing power as a function of frequency, hereafter called a "spectrum". At the edge of the targeted frequency range, the telescope's sensitivity drops away, so the first and last handful of frequencies in any spectrum should always be discarded. When the total power is required, the data as a function of frequency can just be averaged (after discarding the edge channels).

Some of the data for this experiment have been collected in advance of the lab session by a skilled astronomer and will be available for you to analyse. This form of analysis - extracting new information from data taken in the past by other astronomers - is known as archival research. It is important to recognise that this is *not* simulated or artificial data: it has all the defects, experimental uncertainties and potential problems you might associate with data collected yourself. While you are not collecting all the data yourself, you will supplement the archival data with your own measurements, and you will need to understand elements of the telescope's behaviour and control systems in order to interpret the data. Table 1 describes the main commands which are used in their operation and you can look at the SRT operator's manual for details of the telescope and software¹.

- If you are in the lab and seated at a telescope control computer, open the SRT software and explore it. If no telescope is available, this step can be done later, and you should consider the following questions in the context of one of the telescopes currently in use.
- At this point you should **watch the video on the experiment Moodle page** to familiarise yourself with key aspects of the experiment equipment and data. If doing so in the lab, please keep the volume low or use headphones and be aware of the others around you.
- You should also log in to the webcam mounted on the roof of Physics and take a look at the current state of the telescopes. The URL is listed on the whiteboard and on labels on the desk in front of you. It is best entered on the addressbar of a Chrome browser.
- Can you see any potential problems that might affect which objects are visible at any one time (i.e. where the telescopes can point or things that might affect the data

¹<http://www.haystack.mit.edu/edu/undergrad/srt/SRT%20Software/SRTManual.pdf>

Action	Description
Stow	parks the telescope in a standard position at which limit switches are triggered allowing the software to know where the telescope is pointed.
track	toggles whether the telescope tracks the motion of objects across the sky or not.
Azel	points the telescope at a particular location in terms of the Earth-based coordinates of azimuth and elevation (units of degrees).
freq	sets the central frequency in MHz and the mode of frequency sweeping. If possible, mode 4 gives the widest frequency range, but can interfere with other groups. Use mode 1 if this is the case. e.g. type '1420.4 1' to centre on 1420.4 MHz using mode 1.
offset	offset the telescope from its nominal position in azimuth and elevation.
record	record frequency spectrum to disk. Type the file name in the text box at the bottom of the window before clicking on record to specify a name different from the default name.

Table 1: Main commands to control the radio telescope.

collection)? Discuss this with your lab partner. You should each note down your own thoughts in your lab book.

In general you should always record which telescope (i.e. what equipment setup) you are using/looking at, the location of the observatory, the conditions on the day you are observing and any other relevant information (e.g. whether the data you are looking at is affected by radio frequency interference). Make a note of these things now. If you use a different telescope later in the experiment, record the new details. Radio telescopes can be used in sunshine, drizzle or even normal rain, but intense downpours or shaking by strong winds might affect the data.

Two sources are easily visible to these telescopes: the Sun and the Galactic plane. The Sun emits a broad spectrum of frequencies with no obvious features: the power is simply high over the whole spectrum. The Galactic plane emits 21-cm radiation visible as a broad emission line (like a bright line in an optical spectrum) with one or more peaks depending upon exactly where the telescope is pointing.

The radio power of a source is measured in terms of the “brightness temperature”, the temperature of a black-body that produces the same power at the telescope. The brightness temperature T_b scales with the intensity of flux at a given frequency I_ν from a distant source as $I_\nu \propto \nu^2 T_b$. Hence if you’re considering a narrow range of frequencies, power can be considered to vary linearly with brightness temperature².

3.3 Getting started

The telescopes are controlled from software located in a folder called **SRTvsrft** found on the **C:** drive. Once you start the software (file called **srt**), you should see a screen representing the positions of objects on the sky.

²Note though that this is not the same as measured counts so simple Poisson statistics, error $\propto \sqrt{N}$ will not necessarily hold for values of T_b , but will continue to apply for the effect of repeated sampling

Buttons along the top of the window allow you to carry out various actions. The important ones are listed in Table 1; see the SRT operator’s manual for details of the telescope and software³.

Move the telescope from its starting position using the **Azel** option (Table 1) or by clicking on an object in the sky panel. Try not to move outside the dotted rectangle bounding the sky view as this will trigger limit switches which will return the telescope its stow position which is an irritating waste of time. Check the webcam to see that the telescope moves.

Checkpoint 2 - Consult the demonstrator when you have familiarised yourself with the telescope and the goals of the experiment.

3.4 The theoretical beam-width of the telescope

The resolution of ground-based optical telescopes is usually limited by Earth’s atmosphere but the much longer wavelengths of radio astronomy mean that the resolution is normally determined by the diffraction limit:

$$\theta_{\text{FWHM}} \approx 1.22 \frac{\lambda}{D}, \quad (8)$$

where the angle θ_{FWHM} is the Full Width at Half Maximum (FWHM) measured in radians, λ is the wavelength and D is the diameter of the radio dish⁴. The beam-width sets the angular resolution of the experiment. According to the Nyquist sampling criterion (Nyquist, 1928), observations are sufficiently sampled if they are spaced by approximately half the FWHM angle; there is no need to sample more finely than this.

Calculate the beamwidth for the SRT from this theory. Hence establish the optimal angular spacing in degrees that you expect to use for the observations of the Galactic plane in this experiment.

3.5 Measuring the beam-width

The beam-width can be measured experimentally by observing a point source. “Point” here means any source significantly smaller on the sky than the beam-width. The Sun is such a source and by offsetting in a series of steps on each side of the Sun it can be used to measure the beam-width. Once these measurements have been obtained, they may reveal that the telescope is pointing askew of its intended target. In this case they can be used to define offsets in azimuth and elevation which are then applied to make the telescope point more accurately, using the **offset** button in the software.

On the [experiment Moodle page](#) you will find a file of data recorded by the telescope as it is offset by multiples of 1 degree in azimuth and elevation through lines which should pass directly through the Sun. The data file format can be found in Appendix A. **Download** this data and import it into Origin (most people find it easiest to first import the data file into Excel and then copy it to Origin from there). Two of the columns will tell you at what

³<http://www.haystack.mit.edu/edu/undergrad/srt/SRT%20Software/SRTManual.pdf>

⁴More precisely this is the angle between the centre and the first minimum in the diffraction pattern produced by a circular aperture, the so-called “Airy pattern”, but it approximately corresponds to the FWHM as well, and the latter is a more practical measure of the resolution of an optical system.

offset position (i.e. angle from the Sun in Azimuth or Elevation) each measurement was taken. There will likely be several measurements taken at each location which you will often need to average in order to help you deal with any radio frequency interference (i.e. spikes in the data from artificial sources) and to evaluate the uncertainty on the measurements.

At this point you are interested in the average power received by the telescope across the range of frequencies to which it is sensitive (the bandpass). Once you have the data, make plots of signal strength versus frequency to determine which edge bands to reject, and then of total signal strength vs angle from the Sun when stepping in both azimuth and elevation. Measure the FWHM of the profile (it is up to you to work out how to do so) and compare your measurements with the diffraction limit you calculated in the previous section.

Checkpoint 3 - Consult the demonstrator when you have finished analysis of the telescope beam-width.

4 Running the telescopes from a script

Since offsetting the telescope at each step by hand is tedious and prone to human-error, the telescopes are usually controlled with a script (a short computer programme which instructs the control interface without pressing buttons) which is placed in the `C:\vsrt` directory.

Although you will not be running your script for observing the Sun, it is still worth considering how to do so, and noting down an outline script in your lab book that would reproduce the data file you've been given, as described below:

First of all note the lowest elevation limit of your telescope. The radio telescopes are less affected by the Earth's atmosphere than optical telescopes, but pointing right down to the horizon is a bad idea: interference from electronics in Physics and elsewhere in the university can affect your measurement, and your view of the sky may be blocked by trees, landscape or buildings (which radio waves can't penetrate). The telescopes may also have physical limits designed in to their system to prevent this. To ensure that you do not hit the limit switches, you should plan your observations to be at least a few degrees above the horizon and ideally high in the sky. Next look up the current elevation of the Sun at <https://www.esrl.noaa.gov/gmd/grad/solcalc/> (you can use London as a proxy for Coventry or enter the longitude and latitude directly - record which you do) and work out a time when the Sun is higher in the sky than this elevation (ideally close to its maximum elevation for the time of year).

Your command script is a simple text file which should look something like the following:

```
2008:351:11:20:00
: record my_sun_scan
: Sun
:10
: offset 1 0
:10
: offset 2 0
:10
```

The first line is the date and time at which observations should begin (format is year:day:hr:min:sec). The second line starts recording data to an appropriate and informative filename of your

choice and the third line moves the telescope to point to the Sun. The following lines alternate between pausing to take data (:10 means a 10s pause) and moving to a new offset pointing with respect to the Sun (offset in azimuth and elevation in degrees). For further information on command scripts see the SRT user manual. This is effectively a simple programming language and you must be precise with the format.

Look at the data you've been given and decide for yourself what the step size of the offsets was and how many offsets were needed. Why do you think these were selected? Choose an appropriate observation date/time and integration time (pause) at each location. Bear in mind that the cycle time between spectra is 2 or 3 seconds, and that each offset takes a few seconds.

Note down the resulting script (you can skip intermediate az/el steps rather than writing everything out in full as long as the sequence is clear) in your lab book.

5 Measuring the Galactic rotation curve

Now we know the beamwidth of the telescope and should have estimated the optimal spatial sampling (i.e. gap between measurements on the sky) required to take reliable data. We're now going to apply this to the Galactic plane (our edge-on view of the disk of the Milky Way) and tune the central frequency to pick up 21-cm emission. If you haven't done so, make a note of the optimal sampling now and record the range of Galactic longitudes you wish to sample and at what spacing.

Since the hydrogen spin-flip releases photons at a specific wavelength, we see it as an emission line: a peak in surface brightness sitting on top of a background emission level in a spectrum. In some positions our observations might be affected by interference. The Galactic emission should be resolved (i.e. have a broad, well-sampled shape with frequency) whereas interference is usually a narrow feature (one or two adjacent channels with very high flux).

To obtain a rotation curve we need to measure as many points as necessary to sample the beam-width over the Galactic longitude ranges $0 < l < 90^\circ$ and $270 < l < 360^\circ$. How much of these regions we can sample from Coventry depends upon the time of year; you should specify the maximum range available to you in your laboratory notebook by looking at the user interface of the telescope. The Galactic centre itself (Sgr A, $l = 0$) is located at declination -28° and is just beyond observability given our Northerly latitude. We can also take measurements beyond the solar circle ($90 < l < 270^\circ$), although they cannot be used to measure the rotation curve out to $R > R_\odot$ and we haven't provided you with any. What would you expect to see in rotational velocity data as a function of Galactic radius, taken both within and outside the Solar radius, in the light of the theory of section 2?

From the data page for this experiment, download the package of data files, this includes one file for each galactic longitude where a measurement has been provided to you. Compare this to your estimate of the optimal sampling and comment. How many additional measurements do you think you and your lab partner would ideally need to take? Practical considerations and available time mean you should probably aim to analyse between 8 and 10 measurements at different Galactic longitudes during the lab.

You should save the data to disk and then analyse the first file; see appendix A for a description of the file format. Use `Origin` to plot power versus frequency. In order to obtain the value of $V_R(\text{max})$ (or $V_R(\text{min})$, depending upon Galactic longitude), needed to

apply Eq. 6 you need to measure the maximum (or minimum) velocity of emission in one spectrum at each galactic longitude. It is up to you to devise an objective and repeatable method for accomplishing this measurement which you should detail in your notebook. As with many areas of astronomy there isn't necessarily a *right* answer here, but you need to be able to clearly describe what you have chosen to do and why, as well as noting down the measurements at each longitude.

Once you are confident with what the emission line should look like and how to find it in your data, agree with your lab partner which measurements you will take to add to those provided for you. Record this discussion in your lab book.

Checkpoint 4 - Consult the demonstrator when you have considered the first emission line and decided your analysis strategy.

Proceed with data collection and/or analysis of each galactic longitude as required. You may need to modify your plans depending on telescope availability and which parts of the galactic plane are visible.

Record at a minimum the file details, galactic longitude, the minimum (or maximum as appropriate) frequency of the line emission and its uncertainty for each data file, and make note of any additional information you think may be relevant, including diagrams/plots.

The data files give the spectrum as a function of frequency. This can be converted to velocity using Eq. 1, but you must also correct for the motion of the Sun and Earth relative to the “Local Standard of Rest” or LSR which is a point near the Sun which executes a circular orbit around the Galactic centre. You do so by subtracting the reported value of VLSR (actually the component of VLSR which lies along the line of sight) which varies with position on the sky and time of year. It should be recorded in the data file separately for each spectrum (see appendix A), and you should note the values at the beginning and end of an observation.

Checkpoint 5 - Consult the demonstrator when you have completed collection of your measurements and converted them to velocities.

6 Analysis

Use your measurements to derive the rotation curve from Eq. 6 and the constants specified in section 2. Remember to apply the offset from the LSR as described in the previous section, if you haven't done so already, and to calculate uncertainties on your values. Finally use Eq. 7 to plot the mass (in solar masses) enclosed within a radius R as a function of R . Make sure you record your calculations and include clear plots of your results.

Checkpoint 6 - Consult the demonstrator when you have finished plotting your rotation curve and estimate of mass enclosed as a function of galactic radius.

You must now consider the interpretation of your result - this is an important part of this experiment so **leave enough time** to think hard about the following questions and record your thoughts and conclusions:

1. What can you infer about the mass distribution in the galaxy from your results?

2. Does that, or does it not, fit with your theoretical expectations and what are the implications for the assumptions on which those expectations were based?
3. What are the main sources of uncertainty in your analysis? Which are random and which are systematic? How would each type of uncertainty affect your plotted rotation curve and interpretation?
4. What information would be required to measure the entire mass of the Galaxy?
5. Is the rotation curve you've measured consistent with what other researchers have found for the Milky Way or other galaxies? Look in the Library's online database and try an online search for other experimental galaxy rotation curves. The best website for finding astronomy research papers is <https://ui.adsabs.harvard.edu/> but you may need to look at other websites to help you identify the most useful papers.
6. Describe a qualitative or quantitative comparison between your results and those in the literature. Can you suggest the origin of any differences? Remember to record the reference information for the data you're comparing against.

Checkpoint 7 - Consult the demonstrator when you have completed the experiment and/or are ready to leave the laboratory

References

- Carroll, B. and D. Ostlie (2006). *An Introduction to Modern Astrophysics*. Addison Wesley, 2nd ed.
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- Ewen, H. and E. Purcell (1951). Observation of a Line in the Galactic Radio Spectrum: Radiation from Galactic Hydrogen at 1,420 Mc./sec. *Nature* 168, 356.
- Nyquist, H. (1928). Certain topics in telegraph transmission theory. *Trans. AIEE* 47, 617.

A Data file format

The spectra can be recorded to files and then analysed with **Origin**. To do so you need to understand the format in which the data is written. Each spectrum appears on a separate line with the format listed item by item in Table 2 below.

Commands and errors appear on lines beginning with an asterisk '*'. You should aim to plot spectra, remove spectra badly affected by interference and average multiple spectra of the same target.

Item	Comment
Date, time	Colon-separated item consisting of the year, the day number and then the time in hours, minutes and seconds
Azimuth	the azimuth (degrees)
Elevation	the elevation (degrees)
Azimuth offset	the azimuth offset applied with offset (degrees)
Elevation offset	the elevation offset applied with offset (degrees)
Lowest frequency	central frequency of lowest frequency bin (MHz)
Frequency width	frequency width of bins (MHz)
Mode	the frequency scanning mode of freq .
Number of bins	the number of frequency bins, “ nbin ” in the next line.
Signal * nbin	measured brightness temperatures, one for each bin
VLSR	at the end, the correction to get to the LSR is recorded, units of km/s

Table 2: Format of a spectrum line from an SRT data file.

For example (red numbers are the 64 bins of signal data):

```
2023:184:14:04:02 13.0 4.0 0.0 0.0 1420.07 0.00781250 1 64 38.0 48.0 85.0 162.0 327.0 568.0 836.0
1252.0 1809.2 1758.9 1749.7 1687.3 1813.9 1814.2 1766.4 1736.4 1724.2 1712.6 1740.6 1838.8 1729.0
1734.6 1688.6 1745.3 1693.8 1758.1 1710.4 1770.5 1654.5 1701.7 1716.9 1816.6 1689.0 1739.1 1712.0
1726.7 1780.0 1743.8 1747.2 1753.2 1732.3 1699.2 1834.8 1760.9 1765.7 1712.7 1734.7 1768.0 1846.3
1762.0 1801.3 1785.8 1816.9 1746.4 1938.8 1796.1 1826.8 1237.0 888.0 586.0 345.0 170.0 84.0 47.0
vlsr -25.44
```

B Troubleshooting

A problem that you will almost certainly encounter in this experiment is radio interference. Some of this is out of your control, and could be the result of nearby electrical equipment and local radio transmissions. However, there is a problem caused by using multiple radio receivers which can be controlled to some extent, but requires some cooperation between the groups doing the experiment.

The radio detection in this experiment uses what is known as “heterodyning” which requires the generation of a “local oscillator” signal close to the frequency of interest. In the case of the SRTs this signal is 800 kHz higher in frequency than the signal you are looking at. This makes it easier for the electronics but the downside is that potentially each receiver is also a transmitter! For one telescope this is no problem, but with multiple telescopes, it is possible that your local oscillator signal will interfere with the other groups, and vice versa. Therefore you should coordinate with other groups to try to avoid this. As an extra complication, mode '4' of the 'freq' command sweeps across a wider range of frequencies by jumping the local oscillator frequency in three steps, and thus if you use this mode you are cyclically producing 3 interference signals. As a result, mode '1' is to be preferred if other groups are working. This means you have to take care to tune your frequency range to cover the maximum velocity of the 21-cm line.

When not collecting data, all telescopes should be tuned to frequency 1410 mode 1

Other strategies to cope with interference are as follows:

- *Ignore the interference during analysis with Origin:* although the autoscaling of the plots by the data acquisition software means that interference sometimes makes it hard to see the signal of interest, the signal may still be present and easy to measure within **Origin**.
- *Re-tune the central frequency:* if the interference is at one end of the frequency range you are looking at, you might be able to remove it simply by adjusting the range over which you scan in frequency.
- *Wait for a while:* some of the interference is time variable, and it may be that time will cure the problem. This is particularly the case with a saw-toothed distortion that sometimes affects mode 4 observations (but see above about mode 4), but in other cases you might have to wait for quite a while.
- *Look at a different part of the sky:* interference may depend upon the orientation of the telescope. You may be able to return to the original location later and find that the problem has gone.
- *Come out of the software and start again:* this is getting a bit desperate, but might be worth a try in serious cases. Unfortunately, it requires resetting to the **stow** position, costing time.
- *Swapping telescopes with another group:* this is really desperate, but might sometimes be necessary. This would require the other group to have completed their data taking.

UNIVERSITY OF WARWICK
SECOND YEAR PHYSICS LABORATORY
Experiment A3: Radio astronomy
Laboratory Notebook Feedback

While keeping a laboratory notebook for this experiment, you should draw on the experience you have gained to date working in the laboratory, and also bear in mind the information contained in the Notes on keeping a laboratory book. Each student should keep a lab book independently of their lab partner. If you and your lab partner submit identical laboratory notebooks (in part or in full) this will be considered as evidence of potential academic misconduct (collusion and/or plagiarism). This will trigger an investigation and may lead to penalties. You may also find it useful to consider the following points:

1. Have all the learning objectives of the experiment been met? 25%
Have you taken profiles across the Sun? Have you detected and measured HI emission along the Galactic plane? Have you derived a rotation curve?
2. Does the notebook contain a clear and accurate record of the methods used and the experimental results obtained? 25%
Explain **fully** what you did. Supply enough context to make it understandable. Imagine that you return to your book in 10 years time – would **you** understand it? Make sure you describe any difficulties you encountered and what you did to overcome them.
3. Comment on the quality of the data and error analysis. 25%
Error analysis is essential. Don't quote experimental values without some estimate of their uncertainty. It is especially important to understand the level of statistical versus systematic errors. Make sure that logbook shows how you arrived at uncertainty estimates. Use your experience to allow you to make reasoned statements as to what the uncertainties are. Repeated, independent measurements are one of the best ways to estimate statistical uncertainties.
4. Has the student demonstrated a clear understanding of the physical principles underlying the experiment? 25%
Why is the measured width of the Sun larger than its angular diameter? Do you understand how observations of HI emission can be used to derive a rotation curve? Why is only the maximum velocity of emission of interest in this context? Why does the correction to the LSR vary in position and time?